

Group 1: - Topic: Digital Forensic Processes, Standards & Legal Framework

Group Members:

Objective:

To understand the complete digital forensic investigation lifecycle, legal requirements, evidence handling principles, and international forensic standards.

Tasks:

1. Study the digital forensic process lifecycle:
 - Identification
 - Preservation
 - Collection
 - Examination
 - Analysis
 - Reporting
 - Archiving
 - Expert Witness
2. International Standards:
 - ISO/IEC 27037 – Evidence Identification, Collection, Acquisition, Preservation
 - ISO/IEC 27043 – Incident Investigation Principles
 - National Institute of Standards and Technology SP 800-86
 - Scientific Working Group on Digital Evidence Best Practices

Tools:

- Forensic SOP/Manual
- Interpol Digital Forensic Manual
- Flowchart tools

Scenario:

A laptop is seized from a suspect. Explain step-by-step what should be done from seizure to final reporting.

Deliverables:

- 1) Digital Forensic Process flowchart (Describe each step)
- 2) Evidence handling checklist (Mobile, Computer, network, email, etc)
- 3) Chain of custody template

Group 2: - Topic: Computer Forensics

Group Members:

Objective:

To acquire and analyze disk evidence and reconstruct user activity.

Tasks:

1. Acquire forensic image of system partition (C drive).
2. Analyze:
 - Browser history
 - Downloads
 - Recent documents
 - User logon activity
 - Prefetch
 - Recycle Bin
 - LNK files
3. Recover deleted files.

Tools

- Autopsy and FTK Imager

Scenario:

Create a user activity scenario and investigate system artifacts as mentioned in step 2.

Deliverables:

- Steps & screenshots for evidence acquiring
- Evidence screenshots
- Hash values
- Forensic Report

Group 3: - Topic: Mobile device Forensics

Group Members:

Objective:

To extract and analyze mobile device backups for user activity and other artifacts.

Tasks:

- 1) Extract logical image (sample/test phone).
- 2) Analyze:
 - WhatsApp chats
 - Deleted media
 - Call logs
 - Contacts
 - Browser activity
 - Installed applications
 - GPS history

Tools

- I-tune backup for IOS (data extraction)
- ADB backup for android (data extraction)

Scenario:

investigate the user activity to determine what were the last call made, WhatsApp communication, browser activity and any deleted data.

Deliverables:

- Chat/call log and user activity analysis summary
- Media recovery list
- Timeline report
- forensic report

Group 4: - Topic: Network Forensics

Group Members:

Objective:

To analyze packet captures and identify users network activities.

Tasks:

- 1) Analyze PCAP files.
- 2) Identify:
 - DNS requests
 - HTTP/HTTPS traffic
 - File downloads
 - IPs
 - Data exfiltration patterns (create the scenario first)
 - VPN

Scenario:

Capture & investigate an outbound traffic from an internal host (system/network)

Deliverables:

- IOC list
- Traffic summary
- Sessions identified
- forensic report

Group 5: - Topic: Memory (Volatile) Forensics

Group Members:

Objective:

To analyze RAM dumps and identify process activity.

Tasks:

- 1) Analyze RAM dump.
- 2) Identify:
 - Identify all running processes.
 - Select any process
 - Process ID
 - Parent Process ID
 - Process creation time
 - Loaded DLLs
 - Network connections
 - Associated files
 - Threads
 - Memory regions
 - Credentials in memory

Tools

- Autopsy
- FTK Imager

Scenario:

Investigate a selected process (chrome.exe, explorer.exe, powershell.exe) from RAM and determine its execution details

Deliverables:

- Process profile sheet
- Screenshots of commands/output
- Timeline of process activity
- Forensic Report

Group 6: - Topic: Email Forensics

Group Members:

Objective:

To identify phishing, spoofing, and source attribution through email header analysis.

Tasks:

- 1) Analyze:
 - Email headers
 - Sender authenticity/ Message ID
 - SPF/DKIM/DMARC
 - Phishing indicators
 - Signs of spoofing indicators
 - Mail routing path
 - Any embedded artifacts

Tools:

- MXToolbox
- Google Admin Toolbox Messageheader
- VirusTotal
- WHOIS

Scenario:

Analyze multiple phishing/spam emails received in inbox to determine the tasks mentioned above.

Deliverables:

- Header analysis report
- Source IP identification
- Authentication result summary
- IOC list: Ips, Domains, Reply-to addresses, URLs, attachment
- Forensic Report

Group 7: - Topic: Malware Analysis

Group Members:

Objective:

To analyze malware behavior using static and dynamic analysis techniques.

Tasks:

- Static analysis:
 - Hashing
 - Strings extraction
 - PE analysis
 - IP address, Create date & other artifacts
- Dynamic analysis:
 - Behavior observation
 - Registry changes
 - Network connections
 - Mutex
 - Persistence
 - Scheduled tasks

Tools:

- Any.Run, VirusTotal or create your own sandbox

Scenario:

Download any malware sample and Analyze a suspicious executable in isolated lab environment.

Deliverables:

- Malware behavior report
- IOC sheet
- YARA rule (basic)

8. OSINT and Dark Web Investigations

Group Members:

Objective:

To identify digital footprints, exposure, and attribution through open-source intelligence.

Tasks:

- Investigate this WhatsApp Channel:
<https://whatsapp.com/channel/0029Va8LNcnEQlafKWyyet0o>
 - Check for any email addresses
 - Contact details
 - IP Details
 - Location
 - Social media footprints
 - Breach data
 - Achieves data
 - Check footprints in dark web forums
 - Source identification

Tools

Maltego , Shodan or OSINT Framework

Scenario:

Investigate the WhatsApp Channel to identify the source and any leaked data and identify associated exposure artifacts.

Deliverables:

- OSINT Report with findings
- Cover all the tasks

9. USB Forensics Investigation

Group Members:

Objective:

Investigate which users connected USB device with a computer and what files were copied or accessed.

Tasks:

Analyze:

- USB connection history
- Device serial numbers
- First/last connected time
- Mounted volumes
- Drive letters
- File access history
- Shortcut (.lnk) files
- Jump Lists
- Shellbags
- Registry
- Recent files

Tools:

- Autopsy
- Registry Explorer
- FTK Imager

Scenario:

Create the above scenario and suppose a confidential file was leaked. Determine how it accessed, what files who, it was copied to USB.

Deliverables:

- USB timeline
- Device identification
- Files accessed
- Forensic Findings report

10. Windows User Activity Forensics

Group Members:

Objective:

Reconstruct user actions performed on a Windows system.

Tasks:

Analyze:

- Logon/logoff events
- Recently opened files
- Run commands
- USB usage
- Application execution
- UserAssist
- Prefetch
- Recycle Bin
- Event logs

Tools:

- Autopsy
- WinPrefetchView
- Event Viewer
- FTK Imager

Scenario:

Create the scenario and suppose a system is suspected of insider misuse. Reconstruct what the user did in the last 48 hours.”

Deliverables:

- User activity timeline
- Execution history
- Accessed files
- Forensics Report